```
saima@saima-VirtualBox:~$ cd OS-Jackfruit
saima@saima-VirtualBox:~/OS-Jackfruit$ ls
boilerplate  engine.c  project-guide.md  README.md
saima@saima-VirtualBox:~/OS-Jackfruit$
```

```
saima@saima-VirtualBox:/OS-Jackfruit/boilerplate$ cd ~/OS-Jackfruit/boilerplate
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$ nano monitor.c
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$ nano ~/OS-Jackfruit/boilerplate/monito
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./environment-check.sh

== Supervised Runtime Project Preflight ==
[OK] Ubuntu version 24.04 detected.
[OK] No WSL signature detected.
[OK] Virtualized environment detected: oracle.
[WARN] mokutil not installed; cannot auto-check Secure Boot state.
[OK] Kernel headers found at /lib/modules/6.17.0-20-generic/build.
Building user targets and module...
[OK] Boilerplate build succeeded.
[OK] insmod monitor.ko succeeded.
[OK] /dev/container_monitor exists.
[OK] rmmod monitor succeeded.
Preflight passed.
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$
```

```
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$ cd ..
saima@saima-VirtualBox:~/OS-Jackfruit$ mkdir rootfs-base
saima@saima-VirtualBox:~/OS-Jackfruit$ wget https://dl-cdn.alpinelinux.org/alpine
s-3.20.3-x86_64.tar.gz
--2026-04-14 00:59:14--  https://dl-cdn.alpinelinux.org/alpine/v3.20/releases/x86
.gz
Resolving dl-cdn.alpinelinux.org (dl-cdn.alpinelinux.org)... 167.82.50.132, 2a04:
Connecting to dl-cdn.alpinelinux.org (dl-cdn.alpinelinux.org)|167.82.50.132|:443.
HTTP request sent, awaiting response... 200 OK
Length: 3490290 (3.3M) [application/octet-stream]
Saving to: ‘alpine-minirootfs-3.20.3-x86_64.tar.gz’

alpine-miniroot 100%[======>]   3.33M  9.15MB/s    in 0.4s

2026-04-14 00:59:14 (9.15 MB/s) - ‘alpine-minirootfs-3.20.3-x86_64.tar.gz’ saved

saima@saima-VirtualBox:~/OS-Jackfruit$ tar -xzf alpine-minirootfs-3.20.3-x86_64.t
```

```
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$ make
gcc -O2 -Wall -Wextra -o engine engine.c -lpthread
gcc -O2 -Wall -static -o memory_hog memory_hog.c
gcc -O2 -Wall -static -o cpu_hog cpu_hog.c
gcc -O2 -Wall -static -o io_pulse io_pulse.c
make -C /lib/modules/6.17.0-20-generic/build M=/home/saima/OS-Jackfruit/boilerplate modul
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-20-generic'
make[2]: Entering directory '/home/saima/OS-Jackfruit/boilerplate'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.
  You are using:           gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  CC [M]  monitor.o
monitor.c:113:13: warning: ‘kill_process’ defined but not used [-Wunused-function]
  113 | static void kill_process(const char *container_id,
      |             ^~~~~~~~~~~~
monitor.c:98:13: warning: ‘log_soft_limit_event’ defined but not used [-Wunused-function]
   98 | static void log_soft_limit_event(const char *container_id,
      |             ^~~~~~~~~~~~~~~~~~~~
monitor.c:68:13: warning: ‘get_rss_bytes’ defined but not used [-Wunused-function]
   68 | static long get_rss_bytes(pid_t pid)
      |             ^~~~~~~~~~~~~
  MODPOST Module.symvers
  CC [M]  monitor.mod.o
  CC [M]  .module-common.o
  LD [M]  monitor.ko
  BTF [M] monitor.ko
Skipping BTF generation for monitor.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/saima/OS-Jackfruit/boilerplate'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-20-generic'
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$
```

```
Skipping BTF generation for monitor.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/saima/OS-Jackfruit/boilerplate'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-20-generic'
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo insmod monitor.ko
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$ ls -l /dev/container_monitor
crw------- 1 root root 239, 0 Apr 14 01:02 /dev/container_monitor
saima@saima-VirtualBox:~/OS-Jackfruit/boilerplate$
```

```
saima@saima-VirtualBox: ~/OS-Jackfruit
saima@saima-VirtualBox:~/OS-Jackfruit$ make
gcc -o engine engine.c
make -C /lib/modules/6.17.0-20-generic/build M=/home/saima/OS-Jackfruit/boilerplate modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-20-generic'
make[2]: Entering directory '/home/saima/OS-Jackfruit/boilerplate'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  You are using:           gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  CC [M]  monitor.o
monitor.c:113:13: warning: ‘kill_process’ defined but not used [-Wunused-function]
  113 | static void kill_process(const char *container_id,
      |             ^~~~~~~~~~~~
monitor.c:98:13: warning: ‘log_soft_limit_event’ defined but not used [-Wunused-function]
   98 | static void log_soft_limit_event(const char *container_id,
      |             ^~~~~~~~~~~~~~~~~~~~
monitor.c:68:13: warning: ‘get_rss_bytes’ defined but not used [-Wunused-function]
   68 | static long get_rss_bytes(pid_t pid)
      |             ^~~~~~~~~~~~~
  MODPOST Module.symvers
  CC [M]  monitor.mod.o
  CC [M]  .module-common.o
  LD [M]  monitor.ko
  BTF [M] monitor.ko
Skipping BTF generation for monitor.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/saima/OS-Jackfruit/boilerplate'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-20-generic'
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine supervisor ./rootfs-base
[sudo] password for saima:
Sorry, try again.
[sudo] password for saima:
Supervisor started with rootfs: ./rootfs-base
supervisor>
```

```
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-20-generic'
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine supervisor ./rootfs-base
[sudo] password for saima:
Sorry, try again.
[sudo] password for saima:
Supervisor started with rootfs: ./rootfs-base
supervisor> start alpha
Received command: start alpha
supervisor>
```

```
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine start alpha
Started container alpha with PID 12071
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine start beta
Started container beta with PID 12075
saima@saima-VirtualBox:~/OS-Jackfruit$
```

```
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-20-generic'
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine start alpha
[sudo] password for saima:
Sorry, try again.
[sudo] password for saima:
Sorry, try again.
[sudo] password for saima:
Started container alpha with PID 13043
saima@saima-VirtualBox:~/OS-Jackfruit$ cat containers.txt
alpha 13043

saima@saima-VirtualBox:~/OS-Jackfruit$
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine start beta


Started container beta with PID 13065
Container beta running (PID 13065)
saima@saima-VirtualBox:~/OS-Jackfruit$ cat containers.txt
alpha 13043
beta 13065
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine ps
Running containers:
ID: alpha | PID: 13043
ID: beta | PID: 13065
saima@saima-VirtualBox:~/OS-Jackfruit$ S
```

```
supervisor> hello
Received command: hello
supervisor>
```

```
saima@saima-VirtualBox:~/OS-Jackfruit$ cat alpha.log
[alpha] container started
[alpha] running process...
saima@saima-VirtualBox:~/OS-Jackfruit$
```

Captured container logs via logging pipeline

```
[WARNING] alpha exceeded soft memory limit
saima@saima-VirtualBox:~/OS-Jackfruit$
```

```
[KILLED] alpha exceeded hard memory limit
saima@saima-VirtualBox:~/OS-Jackfruit$
```

```
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine start cpu1
Started container cpu1 with PID 13484
Container cpu1 running (PID 13484)
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine start cpu2
Started container cpu2 with PID 13490
Container cpu2 running (PID 13490)
saima@saima-VirtualBox:~/OS-Jackfruit$ sudo ./engine ps
Running containers:
ID: alpha | PID: 13043
ID: beta | PID: 13065
ID: cpu1 | PID: 13484
ID: cpu2 | PID: 13490
saima@saima-VirtualBox:~/OS-Jackfruit$
```

```
saima@saima-VirtualBox:~/OS-Jackfruit$ ps aux | grep engine
saima      13601  0.0  0.0  17820  2376 pts/2    S+   13:22   0:00 grep --color=auto engine
```